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Exercising More Than 150 min/wk After Concussion Is Associated With Sleep Quality Improvements

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Objective: To examine whether a high volume of aerobic exercise after concussion (>150 min/wk) is associated with improved sleep quality over a 1-month period. We hypothesized that more than 150 min/wk of exercise would be associated with improved sleep quality across concussion recovery. **Design:** Prospective cohort observational study. **Setting:** Sports medicine clinic. **Participants:** Adolescents initially tested 8.4 ± 3.5 (range, 2-18) days postconcussion who returned for a follow-up assessment 34.3 ± 7.7 (range: 20-49) days postconcussion. **Main Outcome Measures:** Participants completed the Pittsburgh Sleep Quality Index and the Post-Concussion Symptom Inventory. No specific exercise or sleep recommendations were given beyond what their treating physician provided. Between study visits, participants recorded exercise performed via wrist-worn actigraphy. We calculated average exercise minutes per week and grouped participants as those who exercised more than 150 min/wk versus those who exercised 150 min/wk or less. **Results:** Thirty-six adolescents participated. Fifteen (42%) recorded more than 150 min/wk of aerobic exercise (age = 14.0 ± 1.7 years; 47% female; mean = 5.6 ± 1.2 d/wk of exercise; mean = 49.2 ± 17.5 min/session), and 21 recorded 150 min/wk or less of aerobic exercise (age = 15.0 ± 1.9 years; 76% female; mean = 2.7 ± 1.6 d/wk of exercise; mean = 30.2 ± 7.8 min/session). There were no significant group differences in the proportion of those who self-reported beginning physical activity prior to enrollment (47% vs 33%; $P = .42$) or for initial sleep quality rating (8.0 ± 3.7 vs 8.6 ± 4.1; $P = .67$) or initial concussion symptom severity rating (34.9 ± 28.0 vs 42.6 ± 25.9; $P = .40$). The group that exercised more than 150 min/wk between visits demonstrated significantly greater median PSQI rating improvements than those who exercised 150 min/wk or less, with a large effect size noted (median change [interquartile range] = 5 [3, 7] vs 1 [0, 4]; $P = .008$; Cohen $d = 0.96$). **Conclusion:** Current recommendations suggest that subsymptom aerobic exercise can be beneficial after concussion. Our findings indicate that an exercise volume of more than 150 min/wk led to greater sleep quality improvements than those who exercised below this level. **Key words:** *adolescent*, *mild traumatic brain injury*, *physical activity*, *rest*, *sleep*

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CONCUSSION, defined as a mild traumatic brain injury induced by biomechanical forces,[1] is a challenging injury for clinicians to manage, given the many factors that can influence recovery.[2,3] While effective treatment pathways for patients with concussion have traditionally been difficult to identify,[4–7] an increasing number of studies demonstrate the benefit of early engagement in physical activity and/or aerobic exercise after concussion.[8–10] Several randomized clinical trials reported that individualized early aerobic exercise programs can provide substantial benefit to patients with a concussion, reducing concussion symptom severity and time to symptom resolution.[11–13] While these outcomes are common methods of assessing recovery, concussion can also affect different domains not thoroughly assessed via concussion symptom inventories.

Concussions can induce sleep problems[14] in a substantial number of patients, potentially related to sleep architecture disruptions,[15,16] insomnia,[15,17] or other

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sleep disruptions. Poor sleep can also lead to detrimental postconcussion outcomes, including development of persisting postconcussion symptoms, worse cognitive function, missed school, or a 3 to 4 times longer recovery time.[14,18–24] Adolescents in general may be at risk for delayed circadian rhythms and insufficient sleep independent of a concussion. Among healthy adolescents, poor sleep quality was associated with worse quality of life.[25] Many different potential combinations of biological, psychosocial, environmental, and postinjury behavioral factors may affect adolescents with a concussion. Poor sleep quality after concussion may lead to more severe symptoms,[26,27] and changes in sleep behavior may impact physical activity. While acute exercise appears to be one approach to improve sleep quality,[28] sleep duration may also relate to physical activity on the following day.[29] Furthermore, some concussion symptoms may be a self-limiting factor in physical activity engagement after concussion.[30] Thus, physical activity (eg, general movement throughout the day) and structured exercise (eg, purposeful or structured physical activity) likely interact with sleep health in a bidirectional manner during concussion recovery. Understanding this interaction may provide insights into how treatments may be tailored to improve concussion outcomes.

Postconcussion exercise recommendations within existing studies commonly consist of an exercise intensity below the heart rate that exacerbates ongoing concussion symptoms.[1,11,31] The patient-specific exercise volume (duration and frequency) that best facilitates concussion recovery remains vague.[31] While recommendations of 30 minutes of light daily exercise resulted in similar outcomes over the month after concussion compared with a gradual return to exercise after symptom resolution,[32] other observations indicate a weekly volume of more than 150 min/wk of aerobic exercise was associated with better symptom improvement, lower depression symptoms, and less dizziness than volumes below this level.[13,33] Observing how a high average weekly exercise volume relates to sleep quality after a concussion may provide more specific and pragmatic guidance for clinicians to provide to patients to facilitate recovery.

Our primary purpose was to examine whether a high volume of aerobic exercise after concussion (>150 min/wk) was associated with improved sleep quality over a 1-month period using a prospective observational design. We hypothesized that participants who exercised more than 150 min/wk would demonstrate more sleep quality improvement from initial to follow-up assessments than participants who exercised below this level. Secondarily, we sought to examine whether a total exercise volume (as a continuous variable) predicted sleep quality change across time while adjusting for the independent effects of sex and age, given their influence on postconcussion recovery outcomes.

METHODS

Study design and participants

We conducted a prospective cohort observational study of adolescents between the ages of 10 and 17 years who enrolled between April 2019 and October 2022. Participants were diagnosed with a concussion by one of several board-certified pediatric sports medicine physicians located within a specialty outpatient clinic associated with a tertiary care children's hospital. Concussion diagnoses were defined consistent with the most up-to-date international consensus statement on concussion in sports definition during the study period, as were management recommendations provided to patients.[1] Our inclusion criteria consisted of being 10 to 17 years of age at the time of enrollment, having a Post-Concussion Symptom Inventory (PCSI) score of 9 or more at initial visit to ensure participants had not recovered by the time they enrolled in the study,[26,30,34–36] and beginning activity tracking within 18 days of their injury. Exclusion criteria included those who experienced a lower extremity injury concomitant with their concussion that would affect exercise participation, a concussion within the last 6 months (excluding the index injury), a second head impact before clearance from the initial concussion, documented structural brain injury via neuroimaging (if performed), or a high-velocity injury mechanism (eg, a motor vehicle crash that occurred when traveling at a high speed or a fall from height, determined upon clinical intake prior to recruitment). The study was approved by the local institutional review board, and informed consent/assent was provided by participants and their parents/caregivers before study enrollment.

Following diagnosis, participants enrolled in the study and completed the initial visit within 18 days of concussion (visit 1: mean = 8, SD = 3 days postconcussion; range, 2-18 days). During visit 1, participants completed a set of patient-reported questionnaires and were provided with a wrist-worn activity tracker. They then returned for a follow-up visit (visit 2: mean = 34, SD = 8 days postconcussion; range, 20-49 days) where they completed patient-reported questionnaires. Between visit 1 and visit 2, participants were instructed to wear the provided activity tracker as much as possible and to record any exercise sessions they completed using the device. Participants were not provided with any specific recommendations for exercise habits or sleep beyond what their treating physician provided as standard of care.

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Patient-reported questionnaires

During visit 1, participants completed a demographic and injury history questionnaire, as well as the Pittsburgh Sleep Quality Inventory (PSQI) and the PCSI. Participants repeated the PSQI at visit 2. Within the PSQI, participants were asked to rate 19 items pertaining to sleep, including aspects of subjective sleep quality, efficiency, latency, disturbances, and daytime dysfunction.[37] We calculated a total PSQI rating in line with past work that ranged from 0 to 21, where 0 represented no sleep difficulty across all assessed areas (ie, excellent sleep quality) and 21 represented severe sleep difficulty across all assessed areas (ie, very poor sleep quality).[25,37,38] At visit 1, we asked participants to rate their sleep since their concussion occurred, while at visit 2 we asked participants to rate their sleep since visit 1.[38] In addition to calculating PSQI scores at each visit, we calculated the change in sleep quality between visits by calculating the difference from visit 1 to visit 2. Thus, a positive PSQI rating change represented an improvement in sleep quality from visit 1 to visit 2 (eg, a change value of 6 was calculated when visit 1 PSQI rating was 7 and visit 2 PSQI rating was 1). Accordingly, a negative PSQI rating change represented worsening sleep quality across visits, and a zero PSQI rating change represented no change across visits. We used this PSQI rating change as our primary outcome variable to compare exercise volume between groups so that we could better understand the interaction between exercise volume and sleep assessed at each visit.

Participants also completed the PCSI to rate their overall concussion symptoms.[39,40] Within this concussion symptom severity rating scale, they rated 22 concussion symptoms on a 7-point scale ranging from 0 to 6, where 6 indicated the symptom was a "severe problem" and 0 indicated the symptom was "not a problem." We then calculated the total concussion symptom severity as the sum of all individual symptom ratings, with higher scores represented more severe concussion symptom burden.

Physical activity and exercise tracking

At visit 1, all participants were provided with a wrist-worn activity tracking device (Fitbit Charge 3). They were instructed on use during this visit, including how to manually record exercise sessions on the device. In addition, they were instructed to wear the device as much as possible from the time of visit 1 until they returned for visit 2 and to use the device to record all exercise sessions performed. Our primary grouping variable was based on the average minutes per week they spent performing any exercise. We grouped participants into those who exercised an average of more than 150 min/wk versus those who averaged 150 min/wk or less. This cutoff was chosen in line with past research that showed this level of aerobic exercise was associated with improved concussion symptom severity and mental health outcomes.[13,33]

Following administration of the wrist-worn activity monitors, we used a preregistered account to access and track data quality/participant adherence during study participation. In addition to calculating exercise volume (average minutes per week spent exercising), we were able to calculate adherence as the number of days where the device was worn/the total number of days between visits. If the step count was extremely low on any given day (<300 steps/d), we counted this as a day in which the device was not worn. Prior work has demonstrated that this device is accurate in determining physical activity level.[41] Through the device app, participants chose from more than 20 different exercise modes at the start of each exercise, and if a participant did not manually start an exercise session, the device automatically recorded participant movements after 15 minutes of continuous physical activity with an elevated heart rate. Therefore, all exercise sessions of more than 15 minutes in length were categorized as an exercise bout and the length of the session was recorded (range, 19-102 minutes).

Statistical analysis

Data are presented as mean (SD) or median [interquartile range] for continuous variables and the number within each group (corresponding percentage) for categorical variables. We first compared participant demographic, medical history, injury event, assessment timing, symptom, and adherence characteristics between groups using independent-samples *t* tests for continuous variables and chi-square analyses for categorical variables (or Fisher's exact tests when cell sizes included $n < 5$). To address our primary purpose, we compared PSQI change ratings between groups using an independent-samples *t* test and Cohen's *d* effect size calculations. We assessed effect sizes as small ($d = 0.2$-0.49), moderate ($d = 0.50$-0.79), and large ($d > 0.8$).

At visit 1, we compared PSQI scores between groups (exercised >150 min/wk vs ≤150 min/wk) using an independent-samples *t* test. We also compared PSQI scores between groups at visit 2 to provide descriptive context for our primary outcome variable, PSQI change rating. To account for multiple comparisons, an adjusted *P* value of .025 was used to determine significance in this comparison to account for the 2 comparisons made.

For our secondary purpose, we evaluated whether total exercise volume per week predicted sleep quality change using a multivariable linear regression model. The predictor variable was total exercise volume per week (in hours), and the outcome variable was sleep quality rating (PSQI) change between visits. In this

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model, we used total exercise volume per week as a continuous variable (rather than a binary categorical variable) so that we could identify the relationship between changes in sleep quality across time with the total exercise volume observed. To account for potential confounding variables of preexisting/demographic characteristics, we included variables that differed between groups at a significance level of $P < .1$ as a covariate in the multivariable model. Within this model, we also assessed potential collinearity using variance inflation factors, where a value of less than 2.5 indicated no collinearity was present. All statistical tests were 2-sided, evaluated using a significance level of $\alpha = .05$, and performed using Stata Statistical Software: version 16 (StataCorp, LLC, College Station, Texas).

RESULTS

We enrolled and initially assessed 43 adolescents within 18 days of concussion. Of these, 7 were lost to follow-up and/or did not wear their activity tracking device during the monitoring period. Thus, we included a total of 36 adolescent participants in our analysis. Of these, 15 (42%) recorded an average exercise volume of more than 150 min/wk and were placed in the high exercise volume group, while 21 (58%) were placed in the 150 min/wk or less exercise volume group. The 2 groups demonstrated no significant differences in demographic, medical history, and injury event characteristics (see Table 1). The more than 150 min/wk exercise volume group was slightly younger, had a lower proportion of female adolescents, and those with a history of migraines compared with the 150 min/wk or less exercise volume group (see Table 1); thus, these variables were included as covariates in the multivariable regression model.

The 2 groups were assessed at similar intervals, beginning approximately 1 week postconcussion and returning for a follow-up assessment approximately 5 weeks postconcussion (see Table 2). The 2 groups did not report significant differences in the proportion of those who self-reported beginning physical activity prior to enrollment or in concussion symptom severity ratings at the initial evaluation (see Table 2). There was no significant difference between groups in wrist-worn actigraphy adherence rate or exercise intensity (see Table 2). The total time from concussion to symptom resolution (>150: 23 ± 10 days vs ≤150: 29 ± 18 days; $P = .31$) did not significantly differ between the 2 groups.

The high-volume exercise group demonstrated a significantly greater sleep quality change (improvement) between assessments than the low/moderate exercise volume group, with a large effect size noted (see Figure 1). Thus, an exercise volume of more than 150 min/wk between assessments was associated with a

TABLE 1 *Participant demographics, medical history, and injury event characteristics between the 2 groups*[a]

Variable	Aerobic exercise >150 min/wk (*N* = 15)	Aerobic exercise ≤150 min/wk (*N* = 21)	*P* value
Age, y	14.0 (1.7)	15.0 (1.9)	.09
Sex (female)	7 (47%)	16 (76%)	.09
Height, cm	165.4 (10.0)	163.9 (8.9)	.63
Weight, kg	56.1 (15.3)	61.6 (13.7)	.27
Race/ethnicity			
Native Hawaiian or Other Pacific Islander	0 (0%)	1 (5%)	>.99
Black or African American	1 (7%)	1 (5%)	>.99
White	11 (73%)	12 (57%)	.48
More than one race reported	2 (13%)	7 (33%)	.14
Unknown/not reported	1 (7%)	0 (0%)	.42
Ethnicity (not Hispanic or Latino)	10 (67%)	13 (62%)	.77
Medical/injury history			
Prior concussion	5 (33%)	10 (48%)	.39
Prior migraine diagnosis	0 (0%)	5 (24%)	.06
Preconcussion ADD/ADHD diagnosis	1 (7%)	4 (19%)	.38
Preconcussion anxiety diagnosis	3 (20%)	4 (19%)	.94
Preconcussion depression diagnosis	1 (7%)	6 (29%)	.20
Loss of consciousness at time of injury	4 (27%)	6 (19%)	>.99
Amnesia at time of injury	5 (33%)	5 (24%)	.50
Sports-related concussion	10 (67%)	16 (76%)	.53

Abbreviations: ADD, attention-deficit disorder; ADHD, attention-deficit/hyperactivity disorder.
[a]Data presented as mean (SD) or *n* (% in group).

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TABLE 2 *Comparison of timeline, symptom, and activity tracking adherence characteristics*[a]

Variable	Aerobic exercise >150 min/wk (*N* = 15)	Aerobic exercise ≤150 min/wk (*N* = 21)	*P* value
Assessment timing			
Visit 1 time (days postconcussion)	7.7 (3.8)	8.8 (3.4)	.36
Visit 2 time (days postconcussion)	34.9 (8.7)	33.8 (7.2)	.69
Time between visits (days between visit 1 and visit 2)	27.1 (9.3)	25.0 (7.3)	.44
Initial visit responses			
Reported beginning physical activity between injury and initial visit	7 (47%)	7 (33%)	.42
Symptom severity: Initial visit (PCSI score)	34.9 (28.0)	42.6 (25.9)	.40
Activity/exercise between visits			
Total days wearing activity monitor	17.2 (6.5)	17.6 (6.7)	.87
Adherence rate (% of possible days wearing activity monitor)	89.9 (15.6%)	83.3 (15.5%)	.23
Exercise frequency (average days per week exercising)	5.6 (1.2)	2.7 (1.6)	...
Exercise duration (average minutes per exercise bout)	49.2 (17.5)	30.2 (7.8)	...
Exercise intensity (mean heart rate during exercise)	114.9 (8.8)	117.3 (9.0)	.44

Abbreviation: PCSI, Post-Concussion Symptom Inventory.
[a]Data presented as mean (SD) or *n* (% in group).

greater sleep quality improvement than an exercise volume of 150 min/wk or less. Furthermore, there were no significant between-group differences in total PSQI rating at visit 1 (below a small effect size), but the more than 150 min/wk exercise volume group had significantly lower (better self-reported sleep quality) scores at visit 2 than the 150 min/wk or less exercise volume group, with a large effect size noted (see Figure 2). After adjusting for the independent effects of age and migraine history, a greater exercise volume was significantly associated with a greater improvement on sleep quality ratings across visits (see Table 3). The multivariable model results suggested that for every 1-h/wk increase of exercise, there

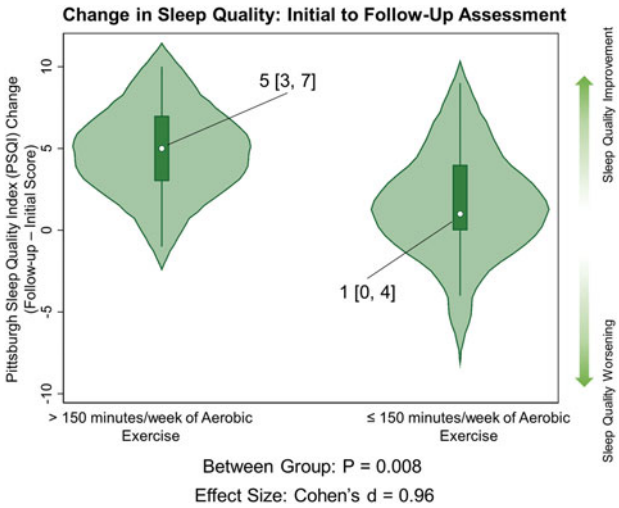

Figure 1. Change in sleep quality (PSQI) from initial to follow-up assessments based on the volume of exercise performed between visits. A positive PSQI change value indicates improved sleep quality at the follow-up assessment relative to the initial postconcussion assessment. Violin plots are presented as median (center dot) and interquartile range (box around the median). The shaded area represents the probability density of data at each level of the scale, smoothed using a kernel density estimator. PSQI indicates Pittsburgh Sleep Quality Index.

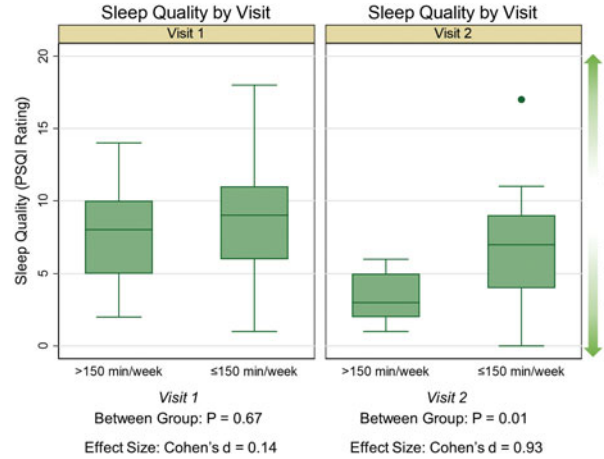

Figure 2. Sleep quality ratings (PSQI) between groups at each visit. The box plot provides each group median, interquartile range, and range during each visit, grouped by those who did and did not exercise for more than 150 min/wk between visits. PSQI indicates Pittsburgh Sleep Quality Index.

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was an expected improvement in sleep quality of 0.82 points on the PSQI. Sex was considered as a covariate but was omitted from the model due to collinearity with exercise volume.

DISCUSSION

The results of our investigation indicate that exercising at a volume that exceeds 150 min/wk after concussion was associated with significant sleep quality improvements across time, with a large effect size noted. In addition, after adjusting for age and preexisting migraine history, a greater exercise volume was associated with more sleep quality improvement across time. While the role of early aerobic exercise for recovery of concussion symptoms remains a key clinical goal, exercise may also have a beneficial role in improving sleep quality following concussion. While our study was observational, and thus causality cannot be determined, this association indicates that sleep quality may interact with exercise volume and represents 2 areas, exercise and sleep, in which interventions can be directed to help optimize recovery following concussion.

The majority of concussion rehabilitation studies have focused on the improvement of overall concussion symptom severity.[9,42,43] To reduce symptoms, many studies have focused on the role of aerobic exercise, largely finding that exercise after a concussion is safe and efficacious.[11,12] While reducing overall concussion symptom severity remains a critical clinical goal, specific mental health or behavioral deficits after concussion remain important patient-centered outcomes for clinical consideration. Prior work has demonstrated the relationship between exercise and postconcussion anxiety, depression, and fear of movement.[33,44] Our data build upon these findings: exercise and sleep quality are associated across time during concussion recovery. In future studies, providing a higher exercise volume recommendation than what is typically recommended (eg, 100 min/wk[12,13]) may allow for patients with concussion to experience better sleep quality and thereby have better quality of life and/or improved function in other domains not thoroughly assessed by concussion symptom inventories alone.[25,38] Sleep problems have traditionally been considered one of many different potential concussion symptoms[1,45,46] rather than a distinct entity that produces additional secondary effects after concussion. Thus, specific intervention efforts aimed toward improving sleep quality may have benefits across many different domains of function and quality of life.

Sleep and exercise likely interact during concussion recovery and plausibly have both independent and combined effects on symptom resolution and/or recovery of different domains of function. While poor sleep is prognostic for poor outcomes (eg, elevated risk of persistent symptoms) and early aerobic exercise tolerance is prognostic for good outcomes (eg, reduced risk of persistent symptoms) after concussion,[11,19,47] few studies have investigated the combined effects of sleep and exercise. Our study results indicate that despite no significant sleep quality differences between groups at visit 1 (small effect size, approximately 1 week postconcussion), those who exercised at a volume that exceeded more than 150 min/wk on average reported better sleep quality at visit 2 (large effect size, approximately 5 weeks postconcussion) than those who had lower exercise volumes. Expanding upon this point, we observed that both groups reported a median value above the threshold indicative of a clinically significant sleep problem (ie, >5)[37] at visit 1, but the median value decreased to below this threshold at visit 2 for the group that exercised more than 150 min/wk only.

We observed that the total exercise volume during the monitoring period (minutes spent exercising per week) predicted the amount of sleep quality change reported between visit 1 and visit 2, after adjusting for patient-specific effects (age, migraine history). Taken together, these associations likely reflect an interaction between sleep and exercise during concussion recovery. However, because of our observational study design, directionality remains unclear. Those who slept better may have felt more refreshed and able to exercise, or exercise may have allowed for a higher energy expenditure during the day,

TABLE 3 *Multivariable regression model results evaluating the association between exercise volume (predictor) with change in PSQI rating between initial/follow-up visits (outcome), while adjusting for the independent effects of age and migraine history*[a]

Variable	β coefficient	SE	*P* value	95% CI
Exercise volume per week, h	.82	0.27	.005	0.26, 1.38
Age, y	.68	0.32	.04	0.02, 1.34
Migraine history	−.73	1.69	.67	−4.17, 2.71

Abbreviation: PSQI, Pittsburgh Sleep Quality Index.
[a]Sex was omitted as a covariate due to collinearity with exercise volume.

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leading to increased drowsiness at night and better sleep. In addition, several factors not investigated in this study could have affected both sleep and exercise participation and may have further modified this relationship. These findings in the context of the well-documented benefits of consistent physical activity and sleep health in the general adolescent population[48–50] indicate that counseling adolescents with a concussion about the role of exercise and sleep during recovery may be worthy of consideration.

The interpretation of our study results should be considered within the limitations of our design. Our findings are associative, and no causal inferences can be made. Sleep and physical activity appear to interact during concussion recovery, but the directionality of this association cannot be determined within our study design. In addition, people who demonstrate different exercise behaviors may have other preexisting health behaviors that influence sleep after concussion, which we did not measure in this study. We also found sleep quality rating changes, but some limitations exist with this measurement approach, namely, the self-reported nature of reporting sleep quality. Among the many postconcussion deleterious effects, sleep problems are commonly reported, but most postconcussion sleep assessments rely on broad, self-report, retrospective questions about sleep. We did not objectively evaluate sleep in this study, and this should be a point of emphasis in future studies. Even with objective evaluation of exercise volume (via wrist-worn actigraphy), some participants may not have worn the device, making it difficult to determine the true nature of postconcussion exercise. Furthermore, sleep health factors affected by concussion include sleep architecture disruptions (sleep fragmentation, decreased REM [rapid eye movement] stage time,[15,16] or insomnia[15,17]) that may persist for months to years after the injury[15,51]; thus, future studies investigating these factors would help inform concussion treatment strategies. Finally, we relied on self-reported diagnosis of preexisting conditions (anxiety, depression, etc), as we were unable to verify true diagnoses before the injury via medical record documentation.

CONCLUSION

Our data indicate that higher exercise volumes during the subacute concussion recovery phase were associated with greater sleep quality improvements across time among adolescents who were primarily seen following a sports-related concussion. Healthcare professionals should consider counseling patients on both sleep health and exercise habits, given that each may play a unique but overlapping role in optimizing their environment to facilitate recovery from concussion.

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